
Algorithm 6 Infer — time-aware bar-phase decoding (inference)

Require : Audio fragment x ; trained θ ; detection threshold τ ;
 candidate meters $\mathcal{M}$ with prior $\hat{\pi}_M(\cdot)$; slip probability ε ;
 maximum advance A ; period window W

Ensure : Predicted events $(\hat{c}_0, \hat{t}_0), \dots, (\hat{c}_{\hat{M}-1}, \hat{t}_{\hat{M}-1})$

Stage 1 : event detection {

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1:    $\{\hat{p}_j(\cdot | x; \theta), \hat{t}_j(\theta, x)\}_{j=0}^{N-1} \leftarrow \text{Forward}(x, \theta)$ 
2:    $\mathcal{J} \leftarrow \{j : 1 - \hat{p}_j(\emptyset | x; \theta) \geq \tau\}$ 
3:    $\hat{M} \leftarrow |\mathcal{J}|$ ; relabel  $j_0 < \dots < j_{\hat{M}-1}$  by increasing  $\hat{t}_j$ 
  }
```

Stage 2 : elapsed beats between consecutive detections {

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4:    $q_i(c) \leftarrow \hat{p}_{j_i}(c) / [\hat{p}_{j_i}(\text{DB}) + \hat{p}_{j_i}(\text{B})]$ ,  $c \in \{\text{DB}, \text{B}\}$ 
5:    $\Delta_i \leftarrow \hat{t}_{j_{i+1}}(\theta, x) - \hat{t}_{j_i}(\theta, x)$ 
6:    $\theta_i \leftarrow \text{median}\{\Delta_k : |k - i| \leq W\}$   $\triangleright$  local beat period
7:    $a_i \leftarrow \text{clip}(\text{round}(\Delta_i / \theta_i), 1, A)$   $\triangleright$  beats elapsed, not assumed to be 1
  }
```

Stage 3 : Viterbi over bar position {

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8:   for  $L \in \mathcal{M}$  do
9:     states  $s \in \{0, \dots, L - 1\}$ , with  $s = 0$  the downbeat
10:     $e_i(s) \leftarrow q_i(\text{DB})$  if  $s = 0$ , else  $q_i(\text{B})$   $\triangleright$  the head cannot resolve  $s > 0$ 
11:    for  $i = 0, \dots, \hat{M} - 2$  do
12:       $T_i[s \rightarrow (s + a_i) \bmod L] \leftarrow 1 - \varepsilon$ 
13:       $T_i[s \rightarrow (s + a_i \pm 1) \bmod L] \leftarrow \varepsilon/2$  each  $\triangleright$  tolerance around  $a_i$ 
14:    end for
15:     $\phi^{(L)}, \ell_L \leftarrow \text{Viterbi}(e, \{T_i\}, p_{\text{init}} = \text{Unif})$ 
16:     $\ell_L \leftarrow \ell_L + \log \hat{\pi}_M(L)$ 
17:  end for
18:   $\hat{L} \leftarrow \arg \max_L \ell_L$ ;  $\phi \leftarrow \phi^{(\hat{L})}$   $\triangleright \phi_0$  is the decoded phase;  $\omega$  is not free
  }
```

19: **for** $i = 0, \dots, \hat{M} - 1$ **do**

20: $\hat{c}_i \leftarrow \text{DB}$ if $\phi_i = 0$, else B; $\hat{t}_i \leftarrow \hat{t}_{j_i}(\theta, x)$ $\triangleright$ event set unchanged from Stage 1

21: **end for**

22: **return** $(\hat{c}_0, \hat{t}_0), \dots, (\hat{c}_{\hat{M}-1}, \hat{t}_{\hat{M}-1})$
